

Linear Algebra

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This is a description of the notes and intended goals/topics that will be covered in them.

Textbooks Used to Develop these Notes:

1. ***Linear Algebra*** - *Hoffman and Kunze*

Other Textbooks Referenced:

1. ***Textbook*** - *Author*

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Contents

1	Canonical Forms and Characteristic Values	3
1.1	Linear Transformations and Coordinates	3
1.2	Characteristic Values	4

NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Canonical Forms and Characteristic Values

1.1 Linear Transformations and Coordinates

We will first recall linear transformations as homomorphisms between two vector spaces. We require the following foundational lemma:

Lemma 1.1. *An n -dimensional vector space V is isomorphic to $\mathbb{F}^n$.*

Proof. Assume β is an ordered basis for V . We claim that the coordinate function $[\]_\beta$ is the linear isomorphism in question.

First, we must prove that it is a linear transformation, such that:

$$[c_1v_1 + c_2v_2]_\beta = c_1[v_1]_\beta + c_2[v_2]_\beta$$

for $v_1, v_2 \in V$. Let $\beta = b_1, \dots, b_n$, then:

$$\begin{aligned} v_1 &= a_1b_1 + \dots + a_nb_n \wedge v_2 = d_1b_1 + \dots + d_nb_n \\ \implies c_1v_1 + c_2v_2 &= (c_1a_1 + c_2d_1)b_1 + \dots + (c_1a_n + c_2d_n)b_n \\ [c_1v_1 + c_2v_2]_\beta &= \begin{bmatrix} c_1a_1 + c_2d_1 \\ \vdots \\ c_1a_n + c_2d_n \end{bmatrix} = \begin{bmatrix} c_1a_1 \\ \vdots \\ c_1a_n \end{bmatrix} + \begin{bmatrix} c_2d_1 \\ \vdots \\ c_2d_n \end{bmatrix} = c_1 \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} + c_2 \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix} = c_1[v_1]_\beta + c_2[v_2]_\beta \end{aligned}$$

To prove injectivity, assume $[v]_\beta = [w]_\beta$, then:

$$[v]_\beta - [w]_\beta = \begin{bmatrix} a_1 - d_1 \\ \vdots \\ a_n - d_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

Thus for each $i \in 1, \dots, n$, we have that $a_i = d_i$, so by the uniqueness of linear combinations for independent vectors, $v = w$.

Finally, surjectivity follows from the fact that:

$$\forall x \in \mathbb{F}^n : x_1b_1 + \dots + x_nb_n \in V$$

by closure under addition and scalar multiplication. Thus $[x_1b_1 + \dots + x_nb_n]_\beta = x$. □

Now consider the linear transformation:

$$T : V \rightarrow W$$

with the ordered basis β_V and β_W . Then for some $v \in V$:

$$v = c_1v_1 + \dots + c_nv_n \implies Tv = c_1Tv_1 + \dots + c_nTv_n$$

Thus $[v]_{\beta_V} = [c_1, \dots, c_n]^T$ and furthermore:

$$[Tv]_{\beta_W} = [c_1Tv_1 + \dots + c_nTv_n]_{\beta_W} = c_1[Tv_1]_{\beta_W} + \dots + c_n[Tv_n]_{\beta_W}$$

Thus, we can represent T as the matrix:

$$[T]_{\beta_V}^{\beta_W} := \begin{bmatrix} [Tv_1]_{\beta_W} & \dots & [Tv_n]_{\beta_W} \end{bmatrix}$$

So, we see that:

$$[Tv]_{\beta_W} = [T]_{\beta_V}^{\beta_W} [v]_{\beta_V}$$

We now explore transformation composition and its interaction with choosing bases. Consider:

$$T : V \rightarrow W \wedge Q : W \rightarrow U$$

with $\beta_V, \beta_W, \beta_U$ being ordered bases for their respective vector spaces. We know by the previous result:

$$[T]_{\beta_V}^{\beta_W} [v]_{\beta_V} = [Tv]_{\beta_W}$$

Then, adding on Q , we have that:

$$\begin{aligned} [Q]_{\beta_W}^{\beta_U} [Tv]_{\beta_W} &= [QTv]_{\beta_U} = [QT]_{\beta_V}^{\beta_U} [v]_{\beta_V} \\ \implies [Q]_{\beta_W}^{\beta_U} \circ [T]_{\beta_V}^{\beta_W} &= [QT]_{\beta_V}^{\beta_U} \end{aligned}$$

We can essentially write this as a commutative diagram. The result allows us to introduce the following:

Definition 1.1 (Similar Matrices). *Two matrices are similar or $A \sim B$ if there exists a matrix P such that:*

$$A = PBP^{-1}$$

Lemma 1.2. *Given a linear operator $T : V \rightarrow V$ (or even a linear transform $T : V \rightarrow W$ where $\dim(V) = \dim(W)$), selecting two ordered bases α, β , we have that:*

$$[T]_{\alpha}^{\alpha} \sim [T]_{\beta}^{\beta}$$

Proof. The logic comes from the idea of change of bases and linear operators. Consider the following commutative diagram and the ordered bases for β, α for V with $T : V \rightarrow V$:

$$\begin{array}{ccccccc} v & \xrightarrow{I} & v & \xrightarrow{T} & Tv & \xrightarrow{I} & Tv \\ \parallel_{\alpha} \downarrow & & \parallel_{\beta} \downarrow & & \downarrow \parallel_{\beta} & & \parallel_{\alpha} \downarrow \\ [v]_{\alpha} & \xrightarrow{[I]_{\alpha}^{\beta}} & [v]_{\beta} & \xrightarrow{[T]_{\beta}^{\beta}} & [Tv]_{\beta} & \xrightarrow{[I]_{\beta}^{\alpha}} & [Tv]_{\alpha} \end{array}$$

Given the fact that $[T]_{\alpha}^{\alpha} [v]_{\alpha} = [Tv]_{\alpha}$, we see that by the commutative diagram:

$$[T]_{\alpha}^{\alpha} = [I]_{\beta}^{\alpha} \circ [T]_{\beta}^{\beta} \circ [I]_{\alpha}^{\beta}$$

We observe the following, applying our results on the composition of multiple linear transforms:

$$[I]_{\beta}^{\alpha} \circ [I]_{\alpha}^{\beta} = [I]_{\alpha}^{\alpha} = I$$

$$[I]_{\alpha}^{\beta} \circ [I]_{\beta}^{\alpha} = [I]_{\beta}^{\beta} = I$$

Thus we can call $P := [I]_{\beta}^{\alpha}$, and we get that $P^{-1} = [I]_{\alpha}^{\beta}$. Then:

$$\begin{aligned} A &:= [T]_{\alpha}^{\alpha} \wedge B := [T]_{\beta}^{\beta} \\ \implies A &= PBP^{-1} \end{aligned}$$

□

1.2 Characteristic Values

For this set of notes, we shall continue to use the terminology of "characteristic values" which readers must note is synonymous to eigenvalues.

Definition 1.2. A linear operator T is simply a linear map from a vector space to itself:

$$T : V \rightarrow V$$

We say that v is a characteristic vector with an associated characteristic value $c \in F$ if:

$$Tv = cv$$

The set of all vectors that follow this property for a certain c is called the characteristic space of c and is equivalently $\ker(cI - T)$.

We already see that if some v is in the characteristic space of c , then its scalar multiples are too:

$$\forall b \in F : T(bv) = b(Tv) = b(cv) = c(bv)$$

We also see that the addition of two vectors in the characteristic space of c is also in the characteristic space:

$$T(v + w) = T(v) + T(w) = cv + cw = c(v + w)$$

The 0 vector is trivially in the characteristic space of any arbitrary c . Thus the characteristic space of c is a subspace of V . If the characteristic space of c is the zero subspace, then it is not a characteristic value.

We will denote the characteristic space of c as E_c from now on.

Lemma 1.3. If 0 is a characteristic value of a linear operator T , then it has a non-trivial kernel.

Proof. This follows trivially as this means there exists some subspace E_0 (which is not the zero subspace), such that:

$$\forall v \in E_0 : Tv = 0v = 0 \implies E_0 \subseteq \ker(T)$$

□

Lemma 1.4. Given a linear transformation $T : V \rightarrow W$:

$$v \in \ker(T)$$

if and only if:

$$[v]_{\beta_V} \in \ker([T]_{\beta_V}^{\beta_W})$$

and $\dim(\ker(T)) = \dim(\ker([T]_{\beta_V}^{\beta_W}))$.

Proof. $\implies$: If $v \in \ker(T)$, then $Tv = 0$, which tells us that:

$$[Tv]_{\beta_W} = [T]_{\beta_V}^{\beta_W} [v]_{\beta_V} = [0]_{\beta_W}$$

$\Leftarrow$: If $[v]_{\beta_V} \in \ker([T]_{\beta_V}^{\beta_W})$, then:

$$[T]_{\beta_V}^{\beta_W} [v]_{\beta_V} = [Tv]_{\beta_W} = [0]_{\beta_W}$$

which by injectivity of the coordinate function implies that $Tv = 0$, so $v \in \ker(T)$. Now let $\mathcal{K}$ be a basis for $\ker(T)$ with vectors $k_1, \dots, k_m$, then we claim that:

$$\mathcal{K}' := \{[k_1]_{\beta_V}, \dots, [k_m]_{\beta_V}\}$$

is a basis for $\ker([T]_{\beta_V}^{\beta_W})$:

$$\forall [v]_{\beta_V} \in \ker([T]_{\beta_V}^{\beta_W}) : [Tv]_{\beta_W} = [0]_{\beta_W}$$

tells us that once again $v \in \ker(T)$, so it can be represented as a linear combination of $\mathcal{K}$ such that:

$$\implies [T]_{\beta_V}^{\beta_W} (c_1[k_1]_{\beta_V} + \cdots + c_m[k_m]_{\beta_V}) = [0]_{\beta_W}$$

We also note that $\mathcal{K}'$ is linearly independent:

$$c_1[k_1]_{\beta_V} + \cdots + c_m[k_m]_{\beta_V} = [0]_{\beta_V} \implies [c_1k_1 + \cdots + c_mk_m]_{\beta_V} = [0]_{\beta_V}$$

And by injectivity of the coordinate function, since $\mathcal{K}$ is linearly independent so that:

$$\implies c_1k_1 + \cdots + c_mk_m = 0 \implies c_1, \dots, c_m = 0$$

Thus $\mathcal{K}'$ is the basis we are looking for, showing the dimensions are equal. $\square$

Theorem 1.1. *The following are equivalent conditions for any arbitrary basis β of the vector space V :*

1. c is a characteristic value of T
2. $cI - T$ is noninvertible
3. $\det(cI - [T]_{\beta}^{\beta}) = 0$

Proof. (1 $\implies$ 2) : If c is a characteristic value, then we have a subspace E_c such that:

$$\forall v \in E_c : Tv = cv \implies (cI - T)v = 0$$

Because $cI - T$ has a nontrivial kernel, it is not invertible (since it is not an injection).

(2 $\implies$ 3) : This follows from determinant properties.

(3 $\implies$ 1) : If $\det(cI - [T]_{\beta}^{\beta}) = 0$, then the kernel is nontrivial, so:

$$\exists v \in \ker(T) : (cI - [T]_{\beta}^{\beta})[v]_{\beta} = 0 \implies [Tv]_{\beta} = [cv]_{\beta}$$

By the injectivity proved in Lemma 1.1, we obtain:

$$Tv = cv$$

which tells us by Definition 2.2, that c is a characteristic value of the operator. $\square$

In order to find the characteristic values, we can use the characteristic polynomial:

Definition 1.3. *The characteristic polynomial for a linear operator T is defined as $q_T(x) := \det(xI - [T]_{\beta}^{\beta})$ (or alternatively $\det(T - xI)$). The roots to such a polynomial are equivalent to the characteristic values by Theorem 1.1.*

Using the theory of similar matrices and determinants, we can also obtain the following result:

Lemma 1.5. *The characteristic polynomial is independent of the chosen basis β of V , despite a different matrix representation.*

Proof. Consider two ordered bases for V , which we call α, β . Then $q_T(x) := \det(xI - [T]_{\beta}^{\beta})$.

By Lemma 2.2, we see that there exists some square matrix P such that:

$$q'_T(x) := \det(xI - [T]_{\alpha}^{\alpha}) = \det(xI - P[T]_{\beta}^{\beta}P^{-1}) = \det(xIPP^{-1} - P[T]_{\beta}^{\beta}P^{-1})$$

$$\begin{aligned}
&= \det((xIP - P[T]_\beta^\beta)P^{-1}) = \det((xPI - P[T]_\beta^\beta)P^{-1}) = \det(PxI - P[T]_\beta^\beta) \det(P^{-1}) \\
&= \det(P) \det(xI - [T]_\beta^\beta) \det(P^{-1}) = \det(xI - [T]_\beta^\beta) = q_T(x)
\end{aligned}$$

Thus the characteristic polynomial is the same. $\square$

Lemma 1.6. *The degree of the characteristic polynomial is $\dim(V)$.*

Proof. This comes from determinant properties, whereby for some $n \times n$ square matrix A :

$$\det(A) := \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n A_{i, \sigma(i)}$$

where S_n is the symmetric group whose elements are permutations σ . If we consider the identity permutation, then we obtain the product:

$$\prod_{i=1}^n (xI - [T]_\beta^\beta)_{i,i} = \prod_{i=1}^n (x - ([T]_\beta^\beta)_{i,i})$$

which is the highest degree term of the characteristic polynomial which is $\dim(V)$ as $[T]_\beta^\beta \in \mathcal{M}_{\dim(V) \times \dim(V)}(F)$. $\square$

We say that a linear operator T is diagonalizable if we can obtain a set of characteristic vectors from the characteristic spaces forms a basis for V :

$$\beta := \{v_1, \dots, v_n\}$$

where $Tv_i = c_i v_i$, telling us that $[Tv_i]_\beta = [c_i v_i]_\beta = c_i [v_i]_\beta = c_i e_i$ by Lemma 2.1:

$$[T]_\beta^\beta = \begin{bmatrix} c_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & c_n \end{bmatrix}$$

Definition 1.4 (Multiplicities). *The algebraic multiplicity of a characteristic value is the number of times it appears as the root of the characteristic polynomial $q_T(x)$. The geometric multiplicity of a characteristic value is the dimension of its characteristic space E_c .*

Lemma 1.7. *The geometric multiplicity of a characteristic value c is always less than or equal to the algebraic multiplicity.*

Proof. $\square$

Definition 1.5. *Given two vector subspaces $V, W \subseteq U$, we define their sum as:*

$$V + W := \{v + w : v \in V \wedge w \in W\}$$

We can write this as a direct sum:

$$V + W = V \oplus W$$

if and only if:

$$\forall v \in V, \forall w \in W : v + w = 0 \implies v, w = 0$$

We call this linear independence of vector subspaces, which can be extended to finite sums of vector subspaces.

We prove a small equivalence of conditions for the two subspace condition, allowing us to expand to the k -subspace condition in Lemma 1.8.

We see that if $0 \subset V \cap W$, then there exists some $v \neq 0$ such that:

$$v \in V \wedge v \in W$$

So take $-v \in V, v \in W$, then $-v, v \neq 0$ but $-v + v = 0$ proving that linear independence implies:

$$V \cap W = 0$$

Now for the converse, if there exists some $v \in V, w \in W$ such that $v + w = 0 \wedge v, w \neq 0$, then $v = -w$.

Since $-w \in V$, then we have that $w \in V \wedge w \in W$, so $0 \subset V \cap W$. We will also require the fact that the sum of subspaces is a subspace, which we can prove easily.

Given $W_1, \dots, W_k$ subspaces, then $\forall j \in 1, \dots, k : 0 \in W_j$, thus:

$$0 \in W_1 + \dots + W_k$$

Now take some $u, v \in W_1 + \dots + W_k$. We see that by definition of subspace addition:

$$\exists w_1, \dots, w_k : u = w_1 + \dots + w_k$$

$$\exists w'_1, \dots, w'_k : v = w'_1 + \dots + w'_k$$

$$\implies u + v = (w_1 + w'_1) + \dots + (w_k + w'_k) \in W_1 + \dots + W_k$$

by subspace property of each W_j . Scalar multiplication follows similarly. We also see that under a direct sum, the representation of a vector v must be a unique combination by linear independence of subspaces.

Lemma 1.8. *The following statements are equivalent regarding direct sums, given subspaces $W_1, \dots, W_k$:*

1. $W_1, \dots, W_k$ are linearly independent

2. The following condition:

$$\forall j \in 1, \dots, k : W_j \cap \sum_{i \neq j} W_i = 0$$

3. Call the bases for each subspace $\beta_1, \dots, \beta_k$, then for:

$$W = W_1 + \dots + W_k$$

$$\beta_W = \beta_1 \cup \dots \cup \beta_k$$

is a valid basis for W .

Proof. (1 $\implies$ 2): If the subspaces are linearly independent, then:

$$\forall w_i \in W_i : w_1 + \dots + w_k = 0 \implies w_1, \dots, w_k = 0$$

Assume there exists some $j \in 1, \dots, k$ such that:

$$0 \subset W_j \cap \sum_{i \neq j} W_i$$

Then we can say there is some $v \neq 0$ such that:

$$v \in W_j \cap \sum_{i \neq j} W_i \implies v \in W_j \wedge -v \in W_j$$

It also tells us that v is in the sum of the W_i subspaces excluding j , so there are some vectors w_i such that:

$$v = \sum_{i \neq j} w_i \wedge w_j = -v \implies \exists w_i \in W_i : w_j + \sum_{i \neq j} w_i = -v + v = 0$$

yet not all of the w_i vectors are the 0 vector.

(2 $\implies$ 3): Take some $w \in W$, then there exists $w_i \in W_i$ for $i \in 1, \dots, k$ such that:

$$w = w_1 + \dots + w_k = (c_1^{(1)}b_1^{(1)} + \dots + c_{m(1)}^{(1)}b_{m(1)}^{(1)}) + \dots + (c_1^{(k)}b_1^{(k)} + \dots + c_{m(k)}^{(k)}b_{m(k)}^{(k)})$$

where $\dim(W_j) = m(j)$ and the superscripts identifying which subspace's bases and coordinates the vectors and scalars represent respectively.

We see that each $w \in W$ can be represented as a linear combination of:

$$\beta_1 \cup \dots \cup \beta_k$$

We now aim to show the basis vectors are linearly independent to each other. If w were the 0 vector in this case, so that:

$$(c_1^{(1)}b_1^{(1)} + \dots + c_{m(1)}^{(1)}b_{m(1)}^{(1)}) + \dots + (c_1^{(k)}b_1^{(k)} + \dots + c_{m(k)}^{(k)}b_{m(k)}^{(k)}) = 0$$

for arbitrary constants in F , then we can prove this iteratively for all $j \in 1, \dots, k$. Consider $j = 1$, then using the results from our mini proof after Definition 1.5:

$$W_1 \cap \sum_{i \neq 1} W_i = 0 \implies (c_1^{(1)}b_1^{(1)} + \dots + c_{m(1)}^{(1)}b_{m(1)}^{(1)}) = 0$$

and

$$(c_1^{(2)}b_1^{(2)} + \dots + c_{m(2)}^{(2)}b_{m(2)}^{(2)}) + \dots + (c_1^{(k)}b_1^{(k)} + \dots + c_{m(k)}^{(k)}b_{m(k)}^{(k)}) = 0$$

By linear independence of β_1 , we have that the first set of constants are equal to 0. We can repeatedly iterate this for all $j \in 2, \dots, k$ and get that the union of bases is a basis for the sum.

(3 $\implies$ 1): Assume $\beta_1 \cup \dots \cup \beta_k$ is a valid basis for W , then:

$$\forall w_1, \dots, w_k : w_1 + \dots + w_k = 0 \implies (c_1^{(1)}b_1^{(1)} + \dots + c_{m(1)}^{(1)}b_{m(1)}^{(1)}) + \dots + (c_1^{(k)}b_1^{(k)} + \dots + c_{m(k)}^{(k)}b_{m(k)}^{(k)}) = 0$$

Because the union of bases is a basis for W , then all the constants are 0 by linear independence, thus all $w_i = 0$, proving linear independence of the subspaces. $\square$

The third equivalent condition will be important for the next theorem. A useful consequence of the third equivalent notion is that:

$$\dim(W) = \dim(W_1) + \dots + \dim(W_k)$$

which comes from the fact that the union of bases is a basis for the sum.

Theorem 1.2. *The following statements are equivalent to diagonalizability:*

1. *The linear operator $T : V \rightarrow V$ is diagonalizable*

2. *If $d_1, \dots, d_k$ are the algebraic multiplicities, then:*

$$q_T(x) = (x - c_1)^{d_1} \dots (x - c_k)^{d_k}$$

and $d_i = \dim(E_{c_i})$ (algebraic multiplicity is equal to geometric multiplicity)

3. The sum of geometric multiplicities adds up to the dimension of the vector space:

$$\dim(V) = \sum_{i=1}^k \dim(E_{c_i})$$

Proof. (1 $\implies$ 2): Assume $d_1, \dots, d_k$ are the algebraic multiplicities of the characteristic values (where $k \leq n$), then given a basis β composed of characteristic vectors, if we order the vectors in a certain way, we get:

$$[T]_{\beta}^{\beta} = \begin{bmatrix} c_1 I_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & c_k I_k \end{bmatrix}$$

where I_j is a $d_j \times d_j$ matrix. If $k = n$, then each $d_i = 1$ and we have c_n characteristic values. If $k < n$, then some of the characteristic values must be shared by some of the vectors, so we can order the basis accordingly.

Then:

$$\det(xI - [T]_{\beta}^{\beta}) = (x - c_1)^{d_1} \cdots (x - c_k)^{d_k}$$

We now must prove that the algebraic multiplicity of each characteristic value is equal to its geometric multiplicity, which we do via Definition 1.2 and Lemma 1.4:

$$E_{c_i} := \ker(c_i I - T) \implies \dim(E_{c_i}) = \dim(\ker(c_i I - T)) = \dim(\ker(c_i I - [T]_{\beta}^{\beta}))$$

Observe that $c_i I - [T]_{\beta}^{\beta}$ (by the uniqueness of the characteristic values) only has d_i columns of 0 vectors with the rest being linearly independent, so:

$$\dim(\ker(c_i I - [T]_{\beta}^{\beta})) = \dim(E_{c_i}) = d_i$$

(2 $\implies$ 3) : For this proof, it suffices to prove that:

$$\dim(V) = \sum_{i=1}^k d_i$$

This follows from Lemma 1.6. We have that $\dim(V) = \deg(q_T(x)) = \sum_{i=1}^k d_i$. Using the second part of statement 2, we obtain:

$$\dim(V) = \sum_{i=1}^k \dim(E_{c_i})$$

(3 $\implies$ 1): We aim to show that:

$$E_{c_1} + \cdots + E_{c_k} = E_{c_1} \oplus \cdots \oplus E_{c_k}$$

We show it via condition 1 in Lemma 1.8. We want to show that:

$$\forall j \in 1, \dots, k : \sum_{i=1}^k v_j = 0 \implies v_j = 0$$

Define:

$$p(x) := \prod_{i=1}^k (c_i - x) \implies p(T) = \prod_{i=1}^k (c_i I - T)$$

In order to proceed with this proof, we need to prove that the factors of $p(T)$ commute. Take an arbitrary c_i, c_j , then:

$$(c_i I - T)(c_j I - T) = c_i c_j I - c_i T - c_j T + T^2 = c_j c_i I - c_j T - c_i T + T^2 = (c_j I - T)(c_i I - T)$$

Observe that we used the commutativity of multiplication in a field F . If this were a module over a ring R , then the factors of $p(T)$ may not commute.

We define:

$$\forall j \in 1, \dots, k : p_j(T) := \prod_{i \neq j} (c_i I - T)$$

So, by commuting factors accordingly, we can eliminate all v_i vectors except v_j :

$$\sum_{i \neq j} p_j(T) v_i = p(T) 0 = 0 \implies p_j(T) v_j = 0$$

We apply v_j to each and obtain:

$$\prod_{i \neq j} (c_i - c_j) v_j = 0 \implies v_j = 0$$

We can do this for all $j \in 1, \dots, k$, implying that all v_j must be zero, showing the characteristic spaces are linearly independent.

As a result, we have a basis for E such that:

$$\beta_E = \beta_1 \cup \dots \cup \beta_k$$

where E is a subspace of V . We also have the consequence that:

$$\dim(E) = \dim(E_{c_1}) + \dots + \dim(E_{c_k}) = \dim(V)$$

Since the dimensions are equal, that means β_E being $\dim(V)$ linearly independent vectors means they span V , thus $V \subseteq E$, proving that $V = E$.

So, we see that $[T]_{\beta_E}^{\beta_E}$ is a basis that diagonalizes the matrix as:

$$\forall j \in 1, \dots, k, \forall b_i^{(j)} \in \beta_E : T b_i^{(j)} = c_j b_i^{(j)}$$

or each of the elements in β_E come from characteristic spaces. □

The important takeaways are that a linear operator is diagonalizable if and only if the algebraic multiplicities of each characteristic value is equal to its geometric multiplicity.

Additionally, it is diagonalizable if and only if the sum of the geometric multiplicities is the dimension of the underlying vector space V . This idea of "annihilating polynomials" will also be important.